QUESTION ONE

(a)

A rectangular prism has dimensions:

- Length = 4x−14x-14x−1
- Width = 2x+52x+52x+5
- Height = x+4x+4x+4

The **total surface area is 1270 cm²**.

Find the value of xxx.

Show all working and justify any algebraic steps.

(b)

The lines

2x+3y=312x+3y=312x+3y=31

and

x2+y2=145x^2+y^2=145x2+y2=145

intersect at two points.

Find the coordinates of both points of intersection.

Clearly show all algebraic working.

(c)

ABCD is a rectangle.

- AB=18AB=18AB=18 m

- BC=10BC=10BC=10 m

Point P lies on DC such that

∠APB=42◦\angle APB=42^\circ∠APB=42◦

Find the perimeter of triangle APB.

Show full reasoning.

(d)

The table below shows a sequence.

x	y
1	17
2	49
3	145
4	433
5	1297

(i) Find an equation for yyy in terms of xxx.

(ii) State FOUR features of the graph.

(iii) Explain whether the graph has an xxx-intercept.

Justify your answer mathematically.

QUESTION TWO

(a)

The angles of a quadrilateral are:

x+10∘, 2x−15∘, 3x+20∘, 4x−35∘x+10^\circ, \; 2x-15^\circ, \; 3x+20^\circ, \; 4x-35^\circx+10∘,2x−15∘,3x+20∘,4x−35∘

Find the value of xxx.

Then determine the size of the largest angle.

(b)

In the diagram below:

- Triangle PQR is right-angled at P.
- PS is perpendicular to the plane containing PQR.
- SQ=12SQ=12SQ=12 cm
- ∠SQP=37∘\angle SQP=37^\circ∠SQP=37∘
- ∠PQR=28∘\angle PQR=28^\circ∠PQR=28∘

Find angle SRQ.

Show all working.

(c)

Factorise completely and solve:

8x3−18x2−35x+45=08x^3-18x^2-35x+45=08x3−18x2−35x+45=0

Give exact answers.

(d)

A swimming pool consists of:

- A rectangular prism section
- A trapezoidal prism section

Dimensions:

- Length = 30 m
- Width = 12 m
- Deep end = 3.2 m
- Shallow end = 1.4 m

Water costs $3.80 per 1000 L.

The pool is filled to 95% capacity.

Calculate the total cost to fill the pool.

Justify each stage of your working.

QUESTION THREE

(a)

A parabola has:

- Turning point (2,12)(2,12)(2,12)
- Passes through (5,−15)(5,-15)(5,−15)

(i) Find the equation of the parabola.

(ii) Find both xxx-intercepts exactly.

(iii) Determine the axis of symmetry and explain its significance.

(b)

A sector has radius rrr.

The central angle is 135∘135^\circ135∘.

Find an exact expression for:

Arc length−Chord length\text{Arc length} - \text{Chord length}Arc length−Chord length

in terms of rrr.

Simplify your answer fully.

(c)

A right-angled triangle has:

- Base =x+2=x+2=x+2
- Height =2x−1=2x-1=2x−1

Area = 117 cm².

The hypotenuse is also known to satisfy:

h=5x2+2x+5h=\sqrt{5x^2+2x+5}h=5x2+2x+5

Determine:

1. The value of xxx
2. The exact hypotenuse
3. Whether the dimensions form a valid right triangle

Justify your conclusion using mathematical reasoning.

EXTREME EXCELLENCE CHALLENGE

The graph of a sequence is defined by

y=a(3)x+by=a(3)^x+by=a(3)x+b

and passes through

(1,10)(1,10)(1,10)

and

(3,82)(3,82)(3,82)

(a)

Find aaa and bbb.

(b)

Determine the equation.

(c)

Find the value of xxx when

y=730y=730y=730

(d)

Explain how each parameter affects:

- the y-intercept
- horizontal asymptote
- growth rate

using this specific equation.